

Zhou Zhanhao

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Education

Xi'an Jiaotong University

B.Eng. in Automation (in progress), School of Automation Science and Engineering

Sep. 2023 – Present

Xi'an, China

- GPA: **3.63/4.3** Average: **89.37/100**
- Core Courses: Fundamentals of Programming (**99**), Digital Image and Video Processing (**98**), Circuits (**95**), Principles of Automatic Control (**90**), Linear Algebra (**89**).

Research Experience

Multi-Behavior Sequential Recommendation

ADM Lab; Advisor: Assistant Professor Qidong Liu

Sep. 2025 – Present

Xi'an Jiaotong University

- Multi-behavior sequential recommendation:** Investigated relationships and complementary signals among clicks, favorites, cart additions, and purchases. Proposed M³SR (*Mamba-Based Multi-Interest Evolution for Multi-Behavior Sequential Recommendation*), with target-behavior-guided calibration (**BGCA**) and multi-interest state evolution (**BI-Mamba**) modules to address behavioral noise and evolving interests. Evaluated on Taobao, IJCAI, and RetailRocket; manuscript submitted to **AAAI 2027** and under review.
- Semantic ID representations:** Explored hierarchical discrete representations of item semantics, encoding items as token sequences and incorporating multi-behavior information to enrich user preference representations.

Project Experience

RoboMaster: Visual Perception and Motion Control

Algorithm Team Lead, Xi'an Jiaotong University RoboMaster Team

Sep. 2023 – Present

- Led perception, motion prediction, and control development for **automatic aiming at rapidly rotating targets**. Combined classical vision and deep learning; addressed detection jitter and sensor synchronization, achieving a competition hit rate above **30%**.
- Localization and mapping:** Optimized the **Point-LIO** front end and integrated **ICP** for local map alignment and pose correction; achieved mapping error below **5 cm** and perception latency below **0.1 s**.
- Planning:** Built grid-based traversability costs with **RPP/DWA** for low-speed avoidance and **MPPI** for dynamic scenarios; designed rule-based route switching and target selection, analyzed latency, and tuned parameters.
- Trajectory control:** Implemented closed-loop **MPC** tracking with lateral deviation and velocity error in a unified cost function, achieving tracking error below **0.1 m**.
- National undergraduate innovation project (2026):** Developed perception and navigation algorithms for heterogeneous inspection robots in dynamic, complex environments.

RoboCup Engineering Challenge: Intelligent Transport Vehicle

Team Member, Provincial Undergraduate Innovation and Entrepreneurship Training Program

2025

- Contributed visual recognition and motion control algorithms for a transport robot; project completed.

Competitions and Honors

2026	RoboMaster University Championship: National 1st / Regional 1st Prize	Algorithm Lead
2026	RoboMaster University League: Regional 1st Prize	Algorithm Lead
2026	Mathematical Contest in Modeling (MCM): Honorable Mention	
2025	RoboMaster Championship: National 2nd / Regional 1st / Infantry 1st Prize	Algorithm Lead
2025	China Robot Competition and RoboCup China Open: 3rd Prize	
2025	National Simulation Innovation Application Competition: Shaanxi 2nd Prize	
2024	Jinyi Scholarship	XJTU
2024	Outstanding Student	XJTU

Leadership and Activities

President, Original Music Club, Xi'an Jiaotong University

Organized original music album releases and campus electronic music festivals; coordinated student musicians and organized participation in the Music Track of the China Collegiate Computing Competition (4C).

Skills and Languages

Programming	Python, C++
Technologies	PyTorch, ROS/ROS2, Point-LIO/FastLIO, ICP, DWA, TEB, MPPI, OpenCV, Linux, Git
English	CET-4: 556 CET-6: 464